

EXPERIMENT - 6

Title

Implement a Program with Basic Commands on Databases and Collections using MongoDB

Aim

To implement and execute **basic MongoDB commands** for:

- Creating databases
 - Creating collections
 - Viewing databases and collections
 - Dropping databases and collections
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Prerequisites

- Basic knowledge of databases
 - MongoDB installed on system
 - MongoDB Compass or MongoDB Shell (mongosh)
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Software Requirements

- OS: Windows / Linux / macOS
 - Database: MongoDB Community Server
 - Tool: MongoDB Shell (mongosh)
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Hardware Requirements

- Processor: Intel i3 or above

- **RAM:** Minimum 4 GB
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Theory

MongoDB is a **NoSQL**, **document-oriented database** that stores data in **JSON-like documents** (BSON).

Important concepts:

- **Database** → Container for collections
- **Collection** → Group of documents
- **Document** → JSON object

MongoDB creates databases and collections **automatically** when data is inserted.

Algorithm

1. Start MongoDB server.
 2. Open MongoDB Shell.
 3. Create and switch database.
 4. Create collections.
 5. Display databases and collections.
 6. Drop collections and database.
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Procedure & Commands

Step 1: Start MongoDB Shell

```
mongosh
```

Step 2: Display Existing Databases

```
show dbs
```

Step 3: Create / Switch Database

```
use foodDeliveryDB
```

→ If database doesn't exist, MongoDB creates it automatically.

Step 4: Check Current Database

```
db
```

Step 5: Create Collection

```
db.createCollection("restaurants")  
db.createCollection("orders")
```

Step 6: Display Collections

```
show collections
```

Step 7: Insert Sample Document

```
db.restaurants.insertOne({  
  name: "Spice Hub",  
  location: "Hyderabad",  
  rating: 4.5  
})
```

Step 8: View Documents

```
db.restaurants.find()
```

Step 9: Drop Collection

```
db.orders.drop()
```

Step 10: Drop Database

```
db.dropDatabase()
```

Sample Output

```
switched to db foodDeliveryDB
restaurants
orders
{
  "_id" : ObjectId("..."),
  "name" : "Spice Hub",
  "location" : "Hyderabad",
  "rating" : 4.5
}
```

Expected Result

The MongoDB database and collections are successfully created, displayed, and deleted using basic MongoDB commands.

Result

Thus, the program using **basic MongoDB database and collection commands** was successfully implemented and executed.

Viva Voce Questions

1. What is MongoDB?
 2. Difference between database and collection?
 3. What is a document?
 4. Is schema required in MongoDB?
 5. What does use dbName do?
 6. When is a database actually created?
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Conclusion

This experiment introduces **core MongoDB concepts** and commands required to manage databases and collections, forming the foundation for CRUD operations.